POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Digital Logic

Semester: Fall

Year : 2024
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe about Digital and Analog System. Differentiate between the Digital and Analog system. 5

   b) Subtract the following using r's complement: 5

      i. $(3085)_{10} - (2040)_{10}$

      ii. $(11011)_2 - (1010)_2$

   c) Show your familiarization to weighted and non-weighted code with example. Write excess 3 code for decimal digits and justify it as a self-complementing code. 5

2. a) Perform the following conversions: 5

      i. $(457)_8 = (?)_{BCD}$

      ii. $(100101)_2 = (?)_{Gray}$

      iii. $(101011.11010)_2 = (?)_8, (?)_{16}$

      iv. $(1010)_{excess\ 3} = (?)_{84\text{-}2\text{-}1}$

   b) What is Universal gate? Draw and explain about all the basic gates that can be derived from NAND gate. 5

   c) Simplify the following expression using Boolean algebra. 5

      i. A'BC+ABC'+A'B'C'+ABC+A'BC'

      ii. (A+B)'.C+(A+BC')+(A'+B)(A+C')

3. a) Using k-map simplify the following Boolean function and implement using Universal gate. 8

   $F = \sum(1,2,3,6,7,9,11,14)$ and $don't\ care\ (d) = \sum(4,10,15)$.

   b) Design a combinational circuit that converts 2421 code to 84-2-1 code. 7

OR

Construct a Full Adder Circuit using two Half Adder and an OR Gate.

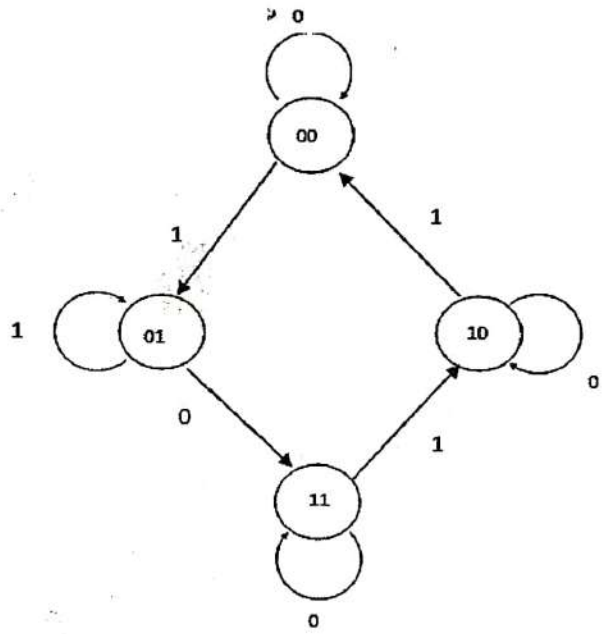
4. a) Why De-Multiplexer is used? Implement F(A,B,C,D) = $\sum$(1,2,4,7,8,10,12,14) using 8 to 1 Multiplexer. 8

OR

Write the logical expression for two bit magnitude comparator and Design a single bit magnitude comparator with all the logical diagrams.

b) A combinational circuit is defined by the function, F1 (A, B, C) = $\sum$ (3, 5, 6, 7), F2 (A, B, C) = $\sum$ (0, 2, 4, 6) implement by using PLA with program table. 7

5. a) From the following state diagram, design a sequential circuit using SR flip flops: 8



b) Differentiate between Flip-flop and Latch. Explain the operation of clocked T flip-flop with the help of its logic diagram, characteristic table, characteristic equation and excitation table. 7

6. a) What is counter? Design a 3 bit asynchronous binary up counter using JK Flip-flop. 8

b) Design Arithmetic Logic Unit (ALU) that performs eight Arithmetic operations and four different logical operations. 7

7. Write short notes on: **(Any two)** 2×5

a) Parity checker

b) Shift Register

c) Processor unit